Lean Business Canvas · Kasi Grid

Amethyst & Azure · FNB App of the Year Hackathon 2026

PROBLEM

- Unscheduled power and water outages in townships with no published, trusted information (load reduction, cable theft, substation faults; 22 major Johannesburg water outages in two weeks of January 2026).
- Households on medical devices cannot tell if a backup will last.
- Small businesses lose stock and sales and cannot see the cost.

SOLUTION

- Live satellite street with verified outages
- Forecast and typical duration
- Medical battery countdown, alerts and ambulance call
- Care and equipment booking
- Outage-cost planner and restock orders
- Community board; accessibility; four languages

UNIQUE VALUE PROPOSITION

The outage app that brings a community closer together: neighbours verify outages, offer help door to door and buy together, so technology connects people instead of isolating them. Free, anonymous, offline, for everyone, anywhere in South Africa.

UNFAIR ADVANTAGE

- Community-generated, device-verified outage history per community that does not exist elsewhere
- Trust in township networks
- A biomedical-engineering view of power dependence
- Works where official data is silent

CUSTOMER SEGMENTS

- Township households (free)
- Medically dependent households and their carers
- Spaza shops and small traders (Pro)
- Suppliers, carers and clinics (partners)
- Municipalities, banks, insurers and NGOs (sponsors)

KEY METRICS

- Verified outages later confirmed (85%)
- Median time to verification (under 20 min)
- Weekly active users and suburbs
- Spoiled stock reduced for spazas (20%)
- Cost per active user (under R5)

CHANNELS

- WhatsApp sharing from the app
- Spaza shop counters (QR code)
- Community groups and ward committees
- Clinics, carers and equipment suppliers
- Installable web app, no app store

COST STRUCTURE

- About R9,800 a month to run a 3,000-user pilot
- About R23,000 one-off (translation review, vetting, POPIA, launch print)
- Moderators, data and outreach are the largest items

REVENUE STREAMS

- Spaza Pro about R49 a month
- Commission on restock orders (4%) and bookings (10%)
- Partner listings about R299 a month
- Municipal dashboard or sponsorship about R3,000 a month

ASSUMPTIONS TO VALIDATE

Hypotheses to validate, not results: spaza willingness to pay, municipal willingness to buy, supplier and carer partnerships, and whether a bank would offer an outage product.

Source code: https://github.com/Unsolvedstar/Kasi-Grid · Live app: https://kasi-grid.vercel.app · Demo video: Watch